

UE24CS352A: Machine Learning

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MACHINE LEARNING

Introduction to Markov Models

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The IID Assumption

- **Independent**:

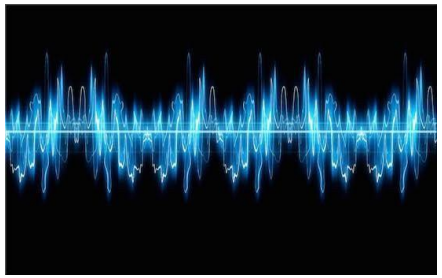
$$P(X \cap Y) = P(X)P(Y)$$

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{P(X)P(Y)}{P(Y)} = P(X)$$

- **Identically Distributed:** drawing from the same prob. distribution

- This assumption does not hold for many real-life scenarios:
  - Any time series data (e.g.: audio/video signal)
  - Probability of words occurring in a sentence
  - Base-pairs in a DNA strand





Stochastic Process

- A collection of random variables indexed by an index set (i.e., time), where all the random variables draw values from a common state space



- Simplest example:
  - Toss a coin N times, to get a vector of length N containing H or T
  - The process is stochastic, where each random variable draws from the state space of H, T

- A Markov process is a specific kind of stochastic process that follows the Markov property:

- A stochastic process {Xt}\{X_t\}{Xt} is said to have the **Markov property** if the **future state depends only on the present state**, and not on the past states.

The Markov Assumption

- Probability of observing the future state depends only on the current state of the process
- In mathematical terms,

$$P(q_n = a|q_1 q_2 \ldots q_{n-1}) = P(q_n = a|q_{n-1})$$

- Where $q_0, q_1, \ldots, q_n$ is a sequence of random variables
- Any sequence of random variables that follows this assumption is said to be a Markov Process of the first order
- For an arbitrary order k we have

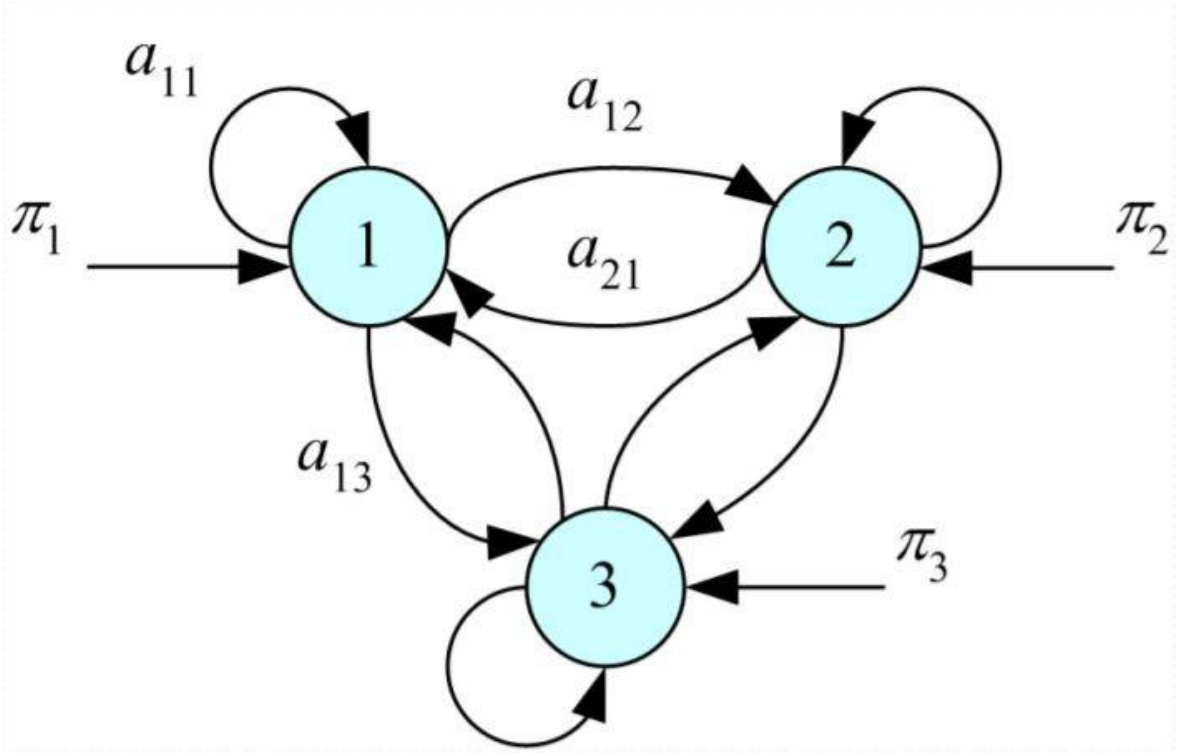
$$P(q_n = a|q_1 q_2 \ldots q_{n-1}) = P(q_n = a|q_{n-1} q_{n-2} \ldots q_{n-k})$$

i.e. The future value depends on the values of the past k states

Markov Chain

- A Markov chain is a discrete-time stochastic process that satisfies the Markov property

- A Markov Chain has the following components:
  - A set of states Q
  - An initial distribution π
  - A transition matrix A(N x N)



Markov Chain: What do the parameters mean?

- π_j : The **initial distribution**, i.e., the probability that the first state outputted by the chain is j

$$\pi_j = P(q_1 = j)$$

- a_{ij}: The **transition probability**, i.e., the probability that the chain will move from state i (at the current time t) to the state j (at the time t+1)

$$a_{ij} = P(q_{t+1} = j \mid q_t = i)$$

Examples of next state prediction given past states

We study stochastic processes because **real-world systems evolve over time under uncertainty** — weather, stock prices, patient health, machine reliability, etc.

So, we naturally want to **predict or understand the future behavior** of such systems.

- In finance, predicting the next market state helps decide whether to buy or sell.
- A self-driving car predicts the next position of nearby vehicles.
- 70% chance of rain tomorrow" tells us about weather persistence.
- In manufacturing, predicting when a machine will fail helps schedule maintenance.
- $P(\text{Rain on Wed} \mid \text{Sunny on Tue})$

  helps in **forecasting**, **resource planning**, and **risk management**.

- "What is the probability the patient will be in a particular health state **tomorrow**, given their current and previous states?"

Example of Likelihood

Observation sequence = daily patient symptom severity:

O=[Healthy, Mild, Mild, Severe, Hospitalized]
HMM trained on historical patients' progression

Compute likelihood P(O/HMM)

Interpretation:

- High likelihood → this progression is **typical**
- Low likelihood → this progression is **unusual**, may need attention

Markov Chain: State Sequence Likelihood

- **Problem Statement:** Given a Markov Chain $\lambda(\pi, A)$ and a sequence of states observed, find the likelihood that the sequence was observed
- **Solution:**

Let the observed state sequence be $Q = \{q_0, q_1, q_2, \dots, q_m\}$

$$P(Q) = P(q_0) \times P(q_1|q_0) \times P(q_2|q_1) \times \dots \times P(q_m|q_{m-1})$$

- $P(q_0)$ can be found from initial distribution π
- $P(q_i|q_j)$ can be found from the matrix A

Markov Chain: Finding Parameters

- The parameters π and A can be found given several state sequences
- The initial distribution π can be found using the below equation, applied to each state:

$$\pi_{q_i} = \frac{\text{Number of sequences that start with state } q_i}{\text{Total Number of sequences given}}$$

- **Important:** sum of all values in π must be 1

- The matrix A can be found using:

$$a_{ij} = \frac{\text{Number of transitions from state } q_i \text{ to state } q_j}{\text{Number of transitions that start from state } q_i}$$

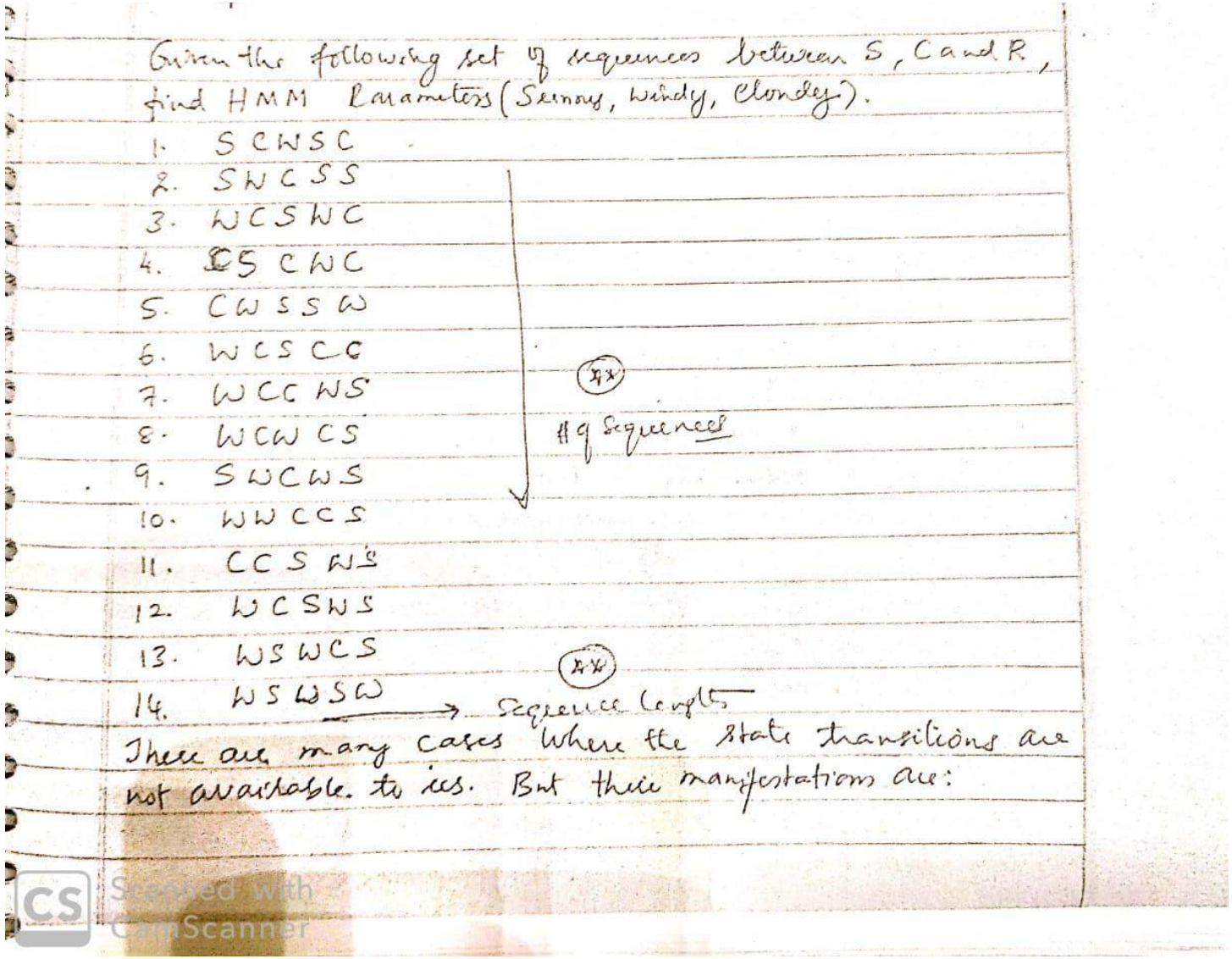
- **Important:** Row sum of A must be 1

MACHINE LEARNING

Markov Chain: Example

States:
Sunny (S),
Cloudy (C),
Windy (W)

Problem Statement: Find the parameters A, π



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Markov Chain: Example Answer

- $\pi = \left\{\frac{3}{14}, \frac{3}{14}, \frac{8}{14}\right\}$ for S, C, W respectively

- $A =$

	Sunny	Cloudy	Windy
Sunny	2/15	4/15	9/15
Cloudy	9/19	4/19	6/19
Windy	9/22	12/22	1/22



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